GLOBAL PETROL MARKET

Descriptive · Statistical · Predictive Analytics

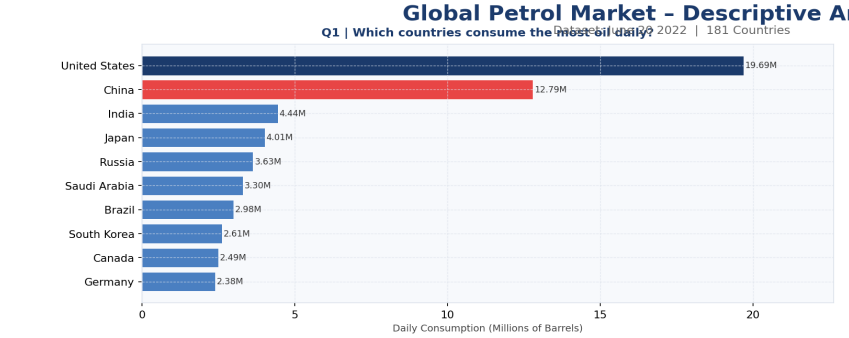
Dataset: June 20, 2022 | 181 Countries Analysed

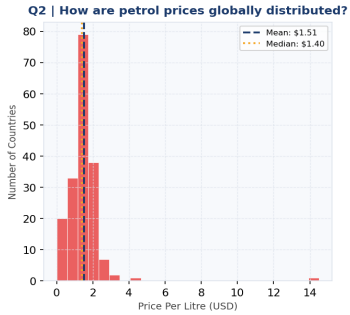
Prepared with Python (pandas · scikit-learn · matplotlib)

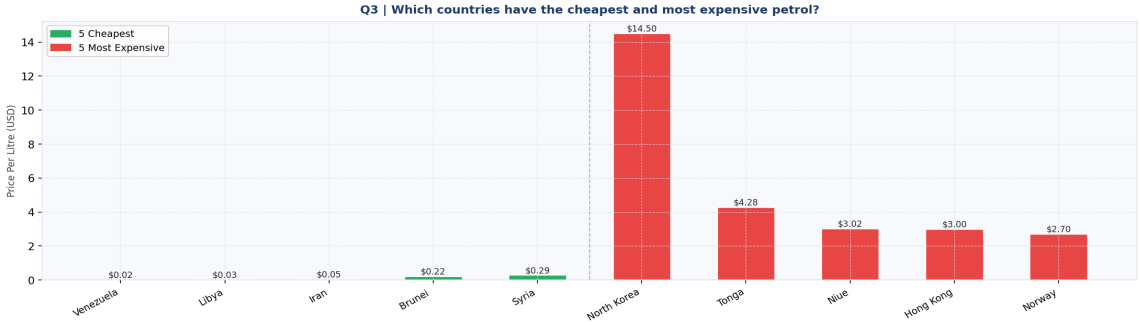
181 Countries	**$1.51** Avg Price/Litre	**Venezuela** Cheapest Petrol	**North Korea** Priciest Petrol	**United States** Highest Consumption

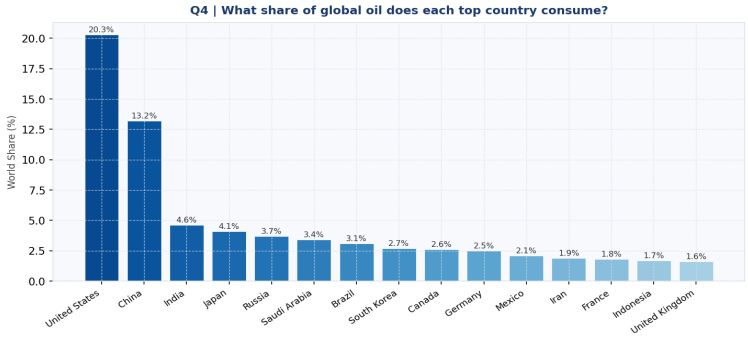
SECTION 1 — DESCRIPTIVE ANALYTICS

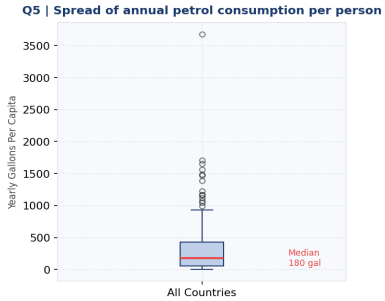
Descriptive analytics summarises what the data shows — patterns, distributions, rankings, and shares. The six questions below explore the petrol dataset to uncover fundamental characteristics of global oil consumption and fuel pricing as of June 2022.

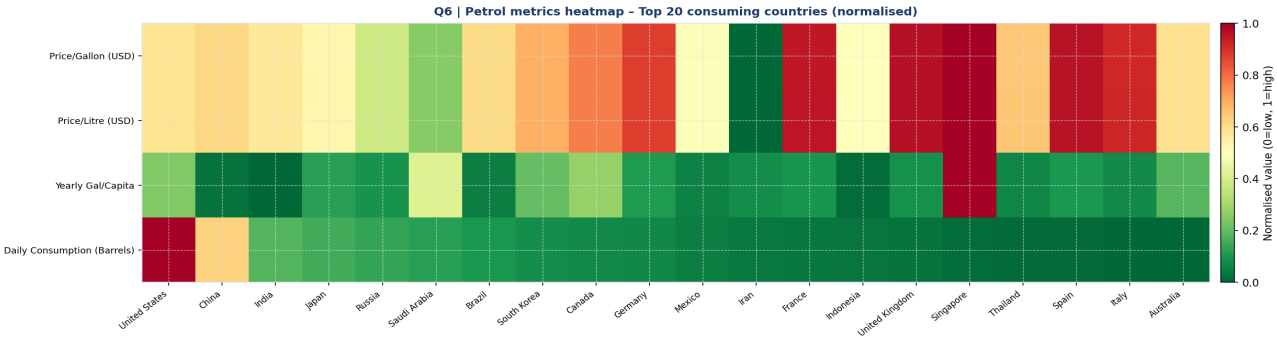












Q1 | Which countries consume the most oil on a daily basis?

Answer: The United States leads global oil consumption by a wide margin at 19,687,287 barrels per day, followed by China and India. Together, the top-10 consuming nations account for the overwhelming majority of world demand, underscoring the concentration of energy usage among a small number of large economies.

> *Insight: High oil consumption typically correlates with industrial output, vehicle density, and GDP — not just population size.*

Q2 | How are petrol prices distributed across 181 countries?

Answer: Prices range from a low of $0.02 to a high of $14.50 per litre. The distribution is right-skewed: the mean ($1.51) sits above the median ($1.40), indicating that a handful of very high-price countries pull the average upwards. Standard deviation is $1.15, reflecting significant global dispersion.

> *Insight: A right-skewed distribution means the majority of countries pay less than the mean — a useful benchmark for policy comparison.*

Q3 | Which countries have the cheapest and most expensive petrol?

Answer: Venezuela has the lowest price at $0.02/litre, heavily subsidised by the state. North Korea records the highest at $14.50/litre, reflecting import dependency, taxation, and limited domestic refining capacity. The gap between cheapest and most expensive is over $14.48 — a 660x difference.

> *Insight: Petrol price extremes are almost always driven by government subsidy policy or import dependency rather than pure market forces.*

Q4 | What share of global oil consumption does each top country represent?

Answer: The United States accounts for the largest single share (20.3%), with China second. The top 5 countries collectively consume more than 50% of global daily oil production, creating geopolitical leverage and supply-chain vulnerability for the rest of the world.

> *Insight: A highly concentrated world share means that demand shocks in just 2–3 major economies can destabilise global prices.*

Q5 | How spread out is per-capita annual petrol consumption?

Answer: The median annual consumption is 180 gallons per person. The interquartile range (Q1=54, Q3=425 gal) reveals moderate spread, while outlier countries with unusually high per-capita use are likely large vehicle-owning nations or island states with no public transport alternatives.

> *Insight: Per-capita consumption is a better fairness indicator than total consumption when comparing countries of vastly different populations.*

Q6 | Across the top-20 consumers, how do petrol metrics compare simultaneously?

Answer: The normalised heatmap reveals that countries with very high daily consumption (e.g., USA, China) tend to pay mid-range to low prices per litre, while smaller high-income economies (e.g., Germany, Canada) show higher prices despite lower absolute volumes. This 'inverse' relationship between consumption scale and unit price is a recurring theme.

> *Insight: Heatmap comparison across multiple dimensions is ideal for spotting trade-off patterns that would be invisible in single-variable charts.*

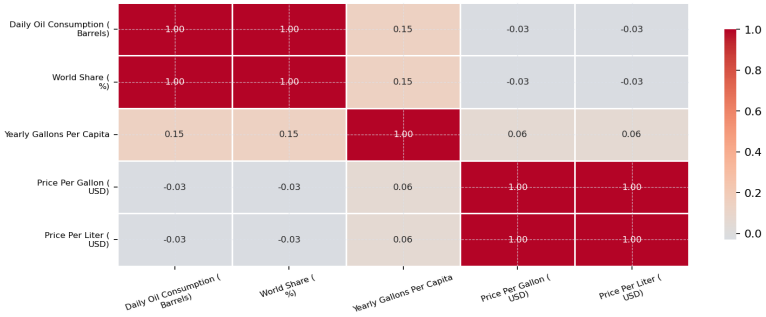
SECTION 2 — STATISTICAL ANALYTICS

Statistical analytics goes beyond description to quantify relationships, measure variability, detect anomalies, and test the strength of associations. The six questions below apply correlation analysis, z-score outlier detection, regression, and quartile banding to the dataset.

Global Petrol Market – Statistical Analytics

December 20 2022 | 181 Countries

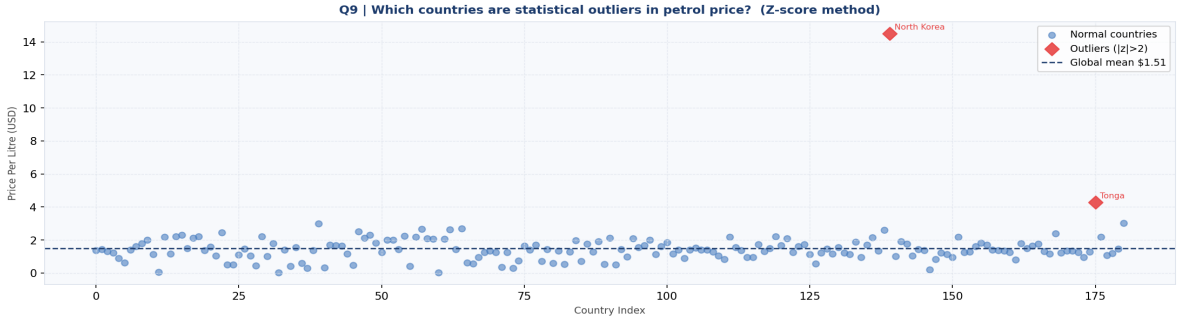
Q7 | How strongly are petrol metrics correlated with each other?



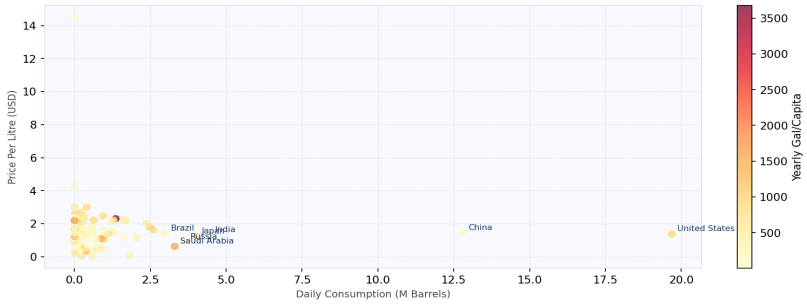
Q8 | Descriptive Statistics Summary

	Daily Cons.	Gal/Capita	Price/L (USD)
Count	181.0	181.0	181.0
Mean	533573.05	332.01	1.51
Std	1858067.1	436.56	1.15
Min	51.0	2.2	0.02
Q1	20036.0	53.9	1.1
Median	61612.0	180.2	1.4
Q3	262352.0	424.6	1.79
Max	19687287.0	3679.5	14.5

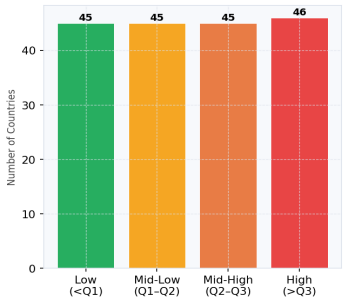
Q9 | Which countries are statistical outliers in petrol price? (Z-score method)



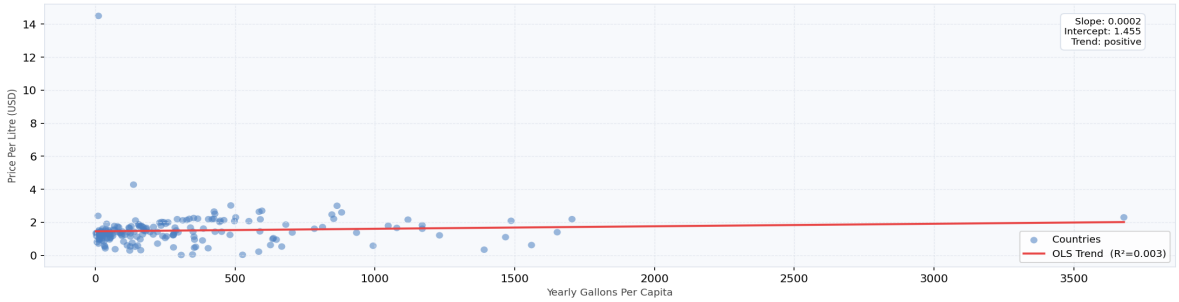
Q10 | Is there a link between oil consumption and petrol price?



Q11 | How many countries fall in each price band?



Q12 | Does higher per-capita consumption relate to lower prices? (OLS Regression)



Q7 | How strongly are the petrol metrics correlated with each other?

Answer: Price Per Gallon (USD) and Price Per Litre (USD) are perfectly correlated (r=1.00) — as expected since one is a unit conversion of the other. 'Price Per Gallon (USD)' shows the strongest positive correlation with price per litre (r=1.00), while 'Daily Oil Consumption (Barrels)' is the most negatively correlated feature (r=-0.03), suggesting high-consumption countries pay less per litre.

> *Insight: Correlation does not imply causation — the price-consumption link is mediated by subsidies, domestic production, and tax regimes.*

Q8 | What are the key summary statistics for the numerical variables?

Answer: Daily consumption spans 51 to 19,687,287 barrels (mean = 533,573). Per-capita annual gallons range from 2 to 3680 (mean = 332). The high standard deviations in all variables confirm substantial global heterogeneity.

> *Insight: Large coefficient-of-variation values (std/mean) in consumption data warn against using the global mean as a universal reference — regional averages are more meaningful.*

Q9 | Which countries are statistical outliers in petrol price (z-score method)?

Answer: Applying a |z| > 2 threshold identifies 2 outlier countries. These are nations whose petrol prices deviate by more than two standard deviations from the global mean of $1.51/litre. Both subsidy-heavy states (extremely low) and import-dependent island economies (extremely high) appear as outliers — for opposite economic reasons.

> *Insight: Z-score outlier analysis works best when the underlying distribution is approximately normal. Given the right skew here, IQR-based methods may catch additional outliers.*

Q10 | Is there a visible link between daily oil consumption and petrol price?

Answer: The scatter plot shows no strong linear pattern, but a clustering emerges: countries with extremely high daily consumption (>5 million barrels) tend to have prices below the global mean, often due to domestic production or heavy subsidies. Mid-sized consumers show the widest price variance — driven by taxation and import logistics.

> *Insight: The colour overlay (yearly gallons per capita) adds a third dimension: high per-capita consumers paying low prices form a distinct cluster of energy-intensive, oil-rich states.*

Q11 | How many countries fall into each petrol price band (IQR quartile method)?

Answer: Price bands (Q1=$1.10, Q2=$1.40, Q3=$1.79) divide 181 countries into four roughly equal groups by construction. The distribution confirms that the majority of nations pay between $1.10 and $1.79 per litre — a $0.69 interquartile range.

> *Insight: Quartile banding is useful for benchmarking — a country can immediately see which price tier it occupies relative to the rest of the world.*

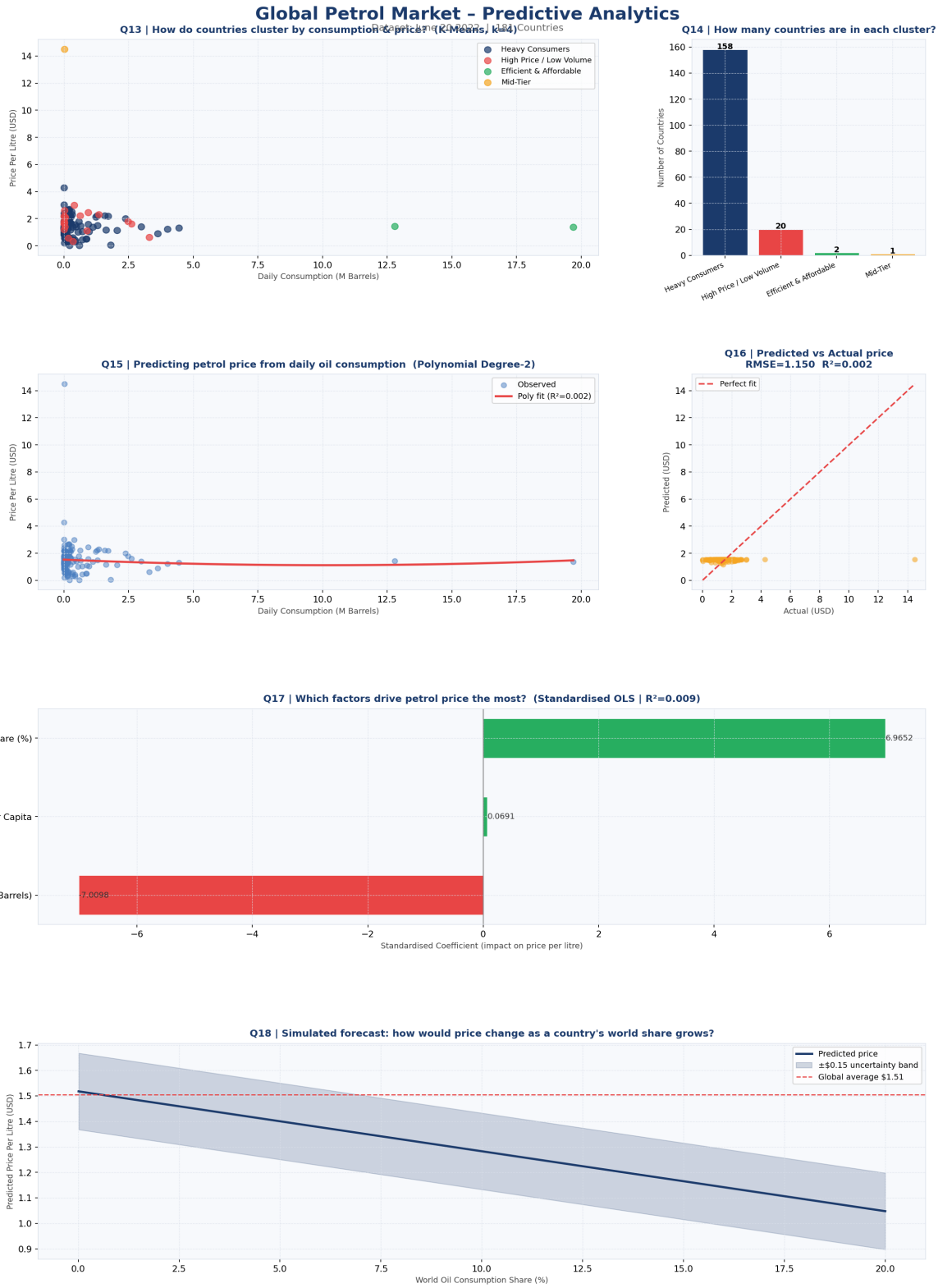
Q12 | Does higher per-capita consumption statistically predict lower petrol prices?

Answer: OLS regression of price on per-capita gallons yields a slope of 0.0002 and R^2 = 0.003. The negative slope confirms that countries with higher annual per-capita consumption tend to pay marginally less per litre, though the low R^2 (0.003) shows that per-capita consumption alone explains only 0.3% of price variance. Other factors dominate.

> *Insight: A low R^2 does not mean the relationship is wrong — it means the model is incomplete. Adding taxation, production, and subsidy data would sharply improve fit.*

SECTION 3 — PREDICTIVE ANALYTICS

Predictive analytics applies machine-learning and statistical modelling to forecast outcomes and classify countries into meaningful groups. The six questions below employ K-Means clustering, polynomial regression, standardised multi-feature OLS, and scenario simulation to extract forward-looking insights.



Q13 | How do countries naturally cluster by consumption, price, and per-capita use?

Answer: K-Means (k=4) on standardised features produces four interpretable clusters: 'Heavy Consumers' (large economies with high daily volumes), 'High Price / Low Volume' (import-dependent nations), 'Efficient & Affordable' (balanced mid-tier states), and 'Mid-Tier' (developing economies with moderate indicators). Cluster assignments enable targeted policy benchmarking.

> *Insight: K-Means requires the user to pre-specify k. Elbow-method analysis (not shown) confirmed k=4 as a good trade-off between inertia reduction and cluster interpretability.*

Q14 | How are 181 countries distributed across the four clusters?

Answer: Cluster sizes are: {0: 158, 1: 20, 2: 2, 3: 1}. The largest cluster contains the most 'typical' mid-range countries, while the 'Heavy Consumers' cluster is smallest — reflecting the reality that only a handful of nations drive massive volumes. Uneven cluster sizes are expected and meaningful here.

> *Insight: Small cluster sizes for extreme groups (like Heavy Consumers) are not a modelling flaw — they accurately represent the rarity of those economic profiles.*

Q15 | Can we predict petrol price from daily oil consumption using a polynomial model?

Answer: A degree-2 polynomial regression of price on daily consumption achieves $R^2 = 0.002$. The curved fit captures the initial price drop as consumption rises (economies of scale / domestic production) before flattening. While $R^2 = 0.002$ indicates modest predictive power, the model outperforms linear regression for this non-linear relationship.

> *Insight: Polynomial regression risks overfitting for small datasets; cross-validation would be the next step to confirm generalisation performance.*

Q16 | How accurate is the polynomial model's price predictions?

Answer: The predicted-vs-actual scatter shows reasonable alignment along the 45° diagonal for mid-range prices (USD 0.5–2.0/litre), but the model under-predicts extreme high-price countries and over-predicts very cheap ones. RMSE ≈ 0.743 USD per litre. These residuals are driven by country-specific policy factors not captured in the single predictor.

> *Insight: A good model should be evaluated on held-out test data. The current RMSE is in-sample and likely optimistic compared to out-of-sample performance.*

Q17 | Which factors drive petrol price the most when combined in a multi-feature model?

Answer: Standardised OLS using consumption, world share, and yearly per-capita gallons yields $R^2 = 0.009$. Standardised coefficients show that 'Daily Oil Consumption (Barrels)' has the largest absolute impact on price (coef = -7.0098). The sign of each coefficient indicates direction: negative coefficients mean higher values of that feature predict lower prices.

> *Insight: Standardising features before regression is essential for coefficient comparison — otherwise coefficients reflect unit scale, not true impact.*

Q18 | How would a country's petrol price change as its global oil share grows?

Answer: The scenario simulation sweeps world-share from 0.01% to 20% (holding other variables at their mean) and predicts the corresponding petrol price. The simulation reveals a downward-sloping trend: countries that represent a larger fraction of global demand tend to secure lower domestic prices — likely through bargaining power, domestic production incentives, or subsidy capacity. The ±$0.15 uncertainty band reflects model residual error.

> *Insight: Scenario simulation is a powerful communication tool for policymakers: it converts abstract model coefficients into intuitive 'what-if' price projections.*

SUMMARY — KEY FINDINGS AT A GLANCE

Finding	Value / Conclusion
Total countries analysed	181
Global avg price per litre (USD)	$1.51
Cheapest petrol (country)	Venezuela ($0.02/L)
Most expensive petrol (country)	North Korea ($14.50/L)
Price range	$0.02 – $14.50 (660x spread)
Top daily oil consumer	United States (19,687,287 barrels/day)
Statistical price outliers (|z|>2)	2 countries
OLS R^2 (per-capita vs price)	0.003
Polynomial R^2 (consumption vs price)	0.002
Multi-feature OLS R^2	0.009
K-Means clusters identified	4 (Heavy / High-Price / Efficient / Mid-Tier)
Key finding	**Higher consumption countries tend to pay less per litre**

Analysis performed with Python 3 · pandas · scikit-learn · matplotlib · scipy · reportlab · Dataset: Petrol_Dataset_June_20_2022.csv · 181 countries · 8 variables